Final Project: Optional part

Due date: The deadline is December 17, 2019, 11:59pm. **The optional part is worth 10 points**. It adds up the the mandatory part. Therefore the maximal grade for the final project is 110.

Goals: The main goal of the project to numerically evaluate the solutions a large class of Hamilton-Jacobi (H-J) partial differential equations in high dimensions using convex minimization algorithms. We will do it for a specific case. This corresponds to implementing a numerical procedure that is described in *Algorithms for overcoming the curse of dimensionality for certain Hamilton–Jacobi equations arising in control theory and elsewhere Research in the Mathematical Sciences.*, J. Darbon and S. Osher, Volume 3, Number 1, September 2016.

We consider $S : \mathbb{R}^n \times [0, +\infty) \to \mathbb{R}$ which satisfies the following Hamilton-Jacobi (H-J) partial differential equation (PDE) with initial data

$$\begin{cases} \dfrac{\partial S}{\partial t}(x,t) + H(\nabla_x S(x,t)) \;=\; 0 & \text{in } \mathbb{R}^n \times (0, +\infty), \\ S(x,0) = J(x) & \text{for any } x \in \mathbb{R}^n. \end{cases}$$

where $\frac{\partial S}{\partial t}$ and $\nabla_x \partial S$ denote the partial derivative. We will assume that the initial data $J : \mathbb{R}^n \to \mathbb{R}$ is convex and that the Hamiltonian $H : \mathbb{R}^n \to \mathbb{R}$ is also convex.

We saw in class that the Fenchel-Legendre $f^* : \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$ transform of a convex function $f : \mathbb{R}^n \to \mathbb{R}$ is defined by

$$f^*(p) = \max_{y \in \mathbb{R}^n} \left(\langle p, y \rangle - f(y) \right)$$

We can represent the (viscosity) solution of the above Hamilton-Jacobi equation as an optimization problem using the Hopf formula which read

$$S(x,t) := \left(J^* + tH\right)^* (x).$$

for any $x \in \mathbb{R}^n$ and any $t > 0$. So we have for any $x \in \mathbb{R}^n$ and any $t > 0$

$$S(x,t) \;=\; -\min_{y \in R} \left\{ J^*(y) - \langle y, x \rangle + tH(y) \right\}$$

In other words, to evaluate the viscosity solution of (H-J) at $x \in \mathbb{R}^n$ and $t > 0$, we need to solve the following minimization problem

$$\min_{y \in \mathbb{R}^n} \left\{ J^*(y) - \langle y, x \rangle + tH(y) \right\}.$$

Let us introduce $f \in \Gamma_0(\mathbb{R}^n)$ and $g \in \Gamma_0(\mathbb{R}^n)$ defined by

$$f(y) \;=\; J^*(y) - \langle y, x\rangle,$$

and

$$g(y) \;=\; t\,H(y).$$

Using these notations we need to solve

$$\min_{y\in R} \{f(y) + g(y)\}\,.$$

We will use the standard alternating direction method of multipliers (ADMM) algorithm (also known as Split-Bregman). It defines sequences $(v^k)_{k\in\mathbb{N}} \subset \mathbb{R}^n$, $(d^k)_{k\in\mathbb{N}} \subset \mathbb{R}^n$ and $(b_k)_{k\in\mathbb{N}} \subset \mathbb{R}^n$ as follows

1. $v^{k+1} \;=\; \arg\min_{v\in\mathbb{R}^n} \left(f(v) + \frac{1}{2}\|d^k - v - b^k\|_2^2 \right)$
2. $d^{k+1} \;=\; \arg\min_{d\in\mathbb{R}^n} \left(g(d) + \frac{1}{2}\|d - v^{k+1} - b^k\|_2^2 \right)$
3. $b^{k+1} \;=\; b^k + v^{k+1} - d^{k+1}$

We set $v^0 = x$, $d^0 = x$ and $b^0 = 0$. Both sequences (v^k) and (d^k) converge to a minimizer of $(f+g)$. We stop the iterations when the following stopping criteria is met:

$$\|v^k - v^{k-1}\|_2^2 \le 10^{-8} \text{ and } \|d^k - d^{k-1}\|_2^2 \le 10^{-8} \text{ and } \|d^k - v^k\|_2^2 \le 10^{-8}.$$

Note that in step 1. we have $f(v) + \frac{1}{2}\|d^k - v - b^k\|_2^2 \;=\; J^*(v) - \langle v, x\rangle + \frac{1}{2}|d^k - v - b^k\|_2^2 \;=\; J^*(v) + \frac{1}{2}\|d^k - b^k + x - v\|_2^2 - \langle d^k - b^k, x\rangle - \frac{1}{2}\|x\|_2^2$. Thus, to compute step 1., we simply need to compute the proximal point of J^* at $(d^k - b^k + x)$.

Here we consider $H(y) = \|y\|_1$ and $J(y) = \frac{1}{2}\|y\|_2^2 - 1$. Therefore we have

$$g(y) = tH(y)$$

By adapting the result we saw in class we have that for any $z \in \mathbb{R}^n$ the minimizer of

$$\min_y \frac{1}{2}\|y - z\|_2^2 + tH(y)$$

which is the proximal point of $g = tH$ at z denoted by $\text{prox}_{tH}(z)$ is for any $i = 1, \ldots, n$

$$\big(\text{prox}_g(z)\big)_i = \begin{cases} z_i - t & \text{if } z_i > t \\ 0 & \text{if } |z_i| \le t \\ z_i + t & \text{if } z_i < -t \end{cases}$$

What you need to do: Now we work with $n = 2$.

1. Give an explicit formula of $J^*(p)$ for any $p \in \mathbb{R}^n$ and show your work (i.e, write down a proof).
2. Give an explicit formula for the proximal point of f, i.e, the minimizer of

$$\frac{1}{2}\|y - z\|_2^2 + f(y)$$

3. Implement ADMM in a matlab.

4. Implement a Matlab function that computes $S(x,t)$ when x and t are given

5. Visualization: We wish to visualize the solutions for $t = 5$. Write a function that computes and stores the values of $S(x_{i,j}, 5)$ for $x_{i,j} = (-20 + \frac{i}{N}40, -20 + \frac{j}{N}40)$ with $N = 256$, $i = 0, \ldots, N$ and $j = 0, \ldots, N$. Plot the function using the matlab function "pcolor" with the option "shading interp" and also plot the level lines using the matlab function "contour". I'll post a matlab code on canvas that performs this plot. Save this plot as an image.

6. Submit your work through canvas along with the mandatory part.